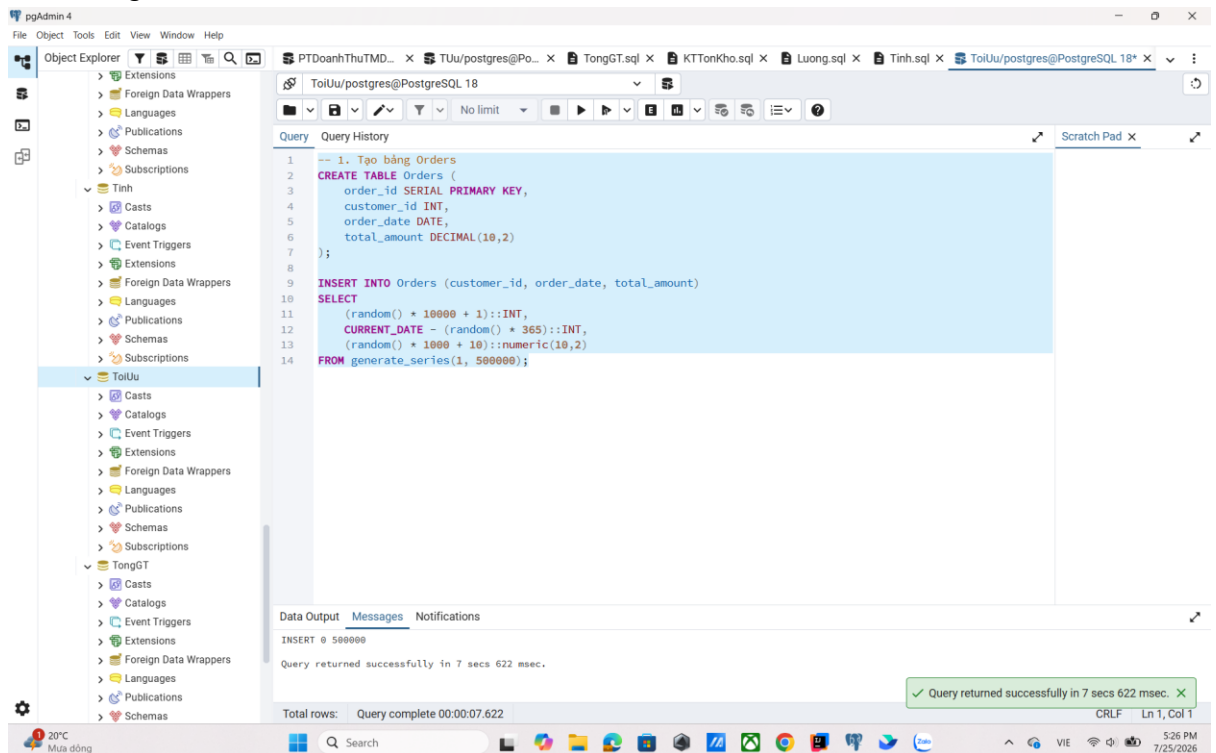


Tên: Nguyễn Hữu Ninh

Lớp: Java Backend (JB-IOC-PTHB260607)

1. Tạo bảng



2. Yêu cầu

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PTDoanhThuTMD... x TUu/postgres@Po... x TongGT.sql x KTTonKho.sql x Luong.sql x Tinh.sql x **ToiUu/postgres@PostgreSQL 18***

ToiUu/postgres@PostgreSQL 18

Query Query History

```
5 order_date DATE,
6 total_amount DECIMAL(10,2)
7 );
8
9 INSERT INTO Orders (customer_id, order_date, total_amount)
10 SELECT
11 (random() * 10000 + 1)::INT,
12 CURRENT_DATE - (random() * 365)::INT,
13 (random() * 1000 + 10)::numeric(10,2)
14 FROM generate_series(1, 500000);
15
16 EXPLAIN ANALYZE
17 SELECT * FROM Orders WHERE customer_id = 500;
```

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

QUERY PLAN
text
1 Gather (cost=1000.00..6794.17 rows=50 width=18) (actual time=1.600..48.759 rows=67.00 loops=1)
2 Workers Planned: 2
3 Workers Launched: 2
4 Buffers: shared hit=3185
5 -> Parallel Seq Scan on orders (cost=0.00..5789.17 rows=21 width=18) (actual time=0.034..4.962 rows=22.33 loops=1)
6 Filter: (customer_id = 500)
7 Rows Removed by Filter: 166644
8 Buffers: shared hit=3185
9 Planning:
10 Buffers: shared hit=28 dirtied=1
11 Planning Time: 0.256 ms
12 Execution Time: 48.786 ms

Total rows: 12 Query complete 00:00:00.077

CRLF Ln 16, Col 1

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PTDoanhThuTMD... x TUu/postgres@Po... x TongGT.sql x KTTonKho.sql x Luong.sql x Tinh.sql x **ToiUu/postgres@PostgreSQL 18***

ToiUu/postgres@PostgreSQL 18

Query Query History

```
7 );
8
9 INSERT INTO Orders (customer_id, order_date, total_amount)
10 SELECT
11 (random() * 10000 + 1)::INT,
12 CURRENT_DATE - (random() * 365)::INT,
13 (random() * 1000 + 10)::numeric(10,2)
14 FROM generate_series(1, 500000);
15
16 EXPLAIN ANALYZE
17 SELECT * FROM Orders WHERE customer_id = 500;
18
19 CREATE INDEX idx_orders_customer_id ON Orders(customer_id);
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 624 msec.

Total rows: Query complete 00:00:00.624

✓ Query returned successfully in 624 msec. ✕

CRLF Ln 19, Col 1

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PTDoanhThuTMD... x TUu/postgres@Po... x TongGT.sql x KTTonKho.sql x Luong.sql x Tinh.sql x **ToiUu/postgres@PostgreSQL 18*** x

ToiUu/postgres@PostgreSQL 18

Query Query History

```
10 SELECT
11     (random() * 10000 + 1)::INT,
12     CURRENT_DATE - (random() * 365)::INT,
13     (random() * 1000 + 10)::numeric(10,2)
14 FROM generate_series(1, 500000);
15
16 EXPLAIN ANALYZE
17 SELECT * FROM Orders WHERE customer_id = 500;
18
19 CREATE INDEX idx_orders_customer_id ON Orders(customer_id);
20
21 EXPLAIN ANALYZE
22 SELECT * FROM Orders WHERE customer_id = 500;
```

Data Output Messages Notifications

Showing rows: 1 to 12 / Page No: 1 of 1

QUERY PLAN
text
1 Bitmap Heap Scan on orders (cost=4.81..186.64 rows=50 width=18) (actual time=0.111..0.180 rows=67 loops=1)
2 Recheck Cond: (customer_id = 500)
3 Heap Blocks: exact=67
4 Buffers: shared hit=67 read=3
5 -> Bitmap Index Scan on idx_orders_customer_id (cost=0.00..4.80 rows=50 width=0) (actual time=0.098..0.098 rows=67 loops=1)
6 Index Cond: (customer_id = 500)
7 Index Searches: 1
8 Buffers: shared read=3
9 Planning:
10 Buffers: shared hit=18 read=1
11 Planning Time: 0.252 ms
12 Execution Time: 0.194 ms

Total rows: 12 Query complete 00:00:00.044

Successfully run. Total query runtime: 44 msec. 12 rows affected. X

CRLF Ln 21, Col 1

20°C Mưa dông

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